



- 3 A traffic expert claims that the number of breakdowns occurring each day on a busy section of a motorway follows a Poisson distribution with mean 0.7. The number of breakdowns each day over a 200-day period was recorded. The following table contains the observed frequencies together with some of the expected frequencies using the expert's distribution.

|                              |        |     |        |       |       |          |
|------------------------------|--------|-----|--------|-------|-------|----------|
| Number of breakdowns per day | 0      | 1   | 2      | 3     | 4     | $\geq 5$ |
| Observed frequency           | 88     | 73  | 26     | 7     | 3     | 3        |
| Expected frequency           | 99.317 | $m$ | 24.333 | 5.678 | 0.994 | $n$      |

- (a) Find the value of  $m$  and the value of  $n$ , correct to 3 decimal places. [2]

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- (b)** Carry out a goodness of fit test at the 5% significance level to investigate the expert's claim. [6]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, which is otherwise blank white space.

